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Agriculture in Southeast Asia

Ana Doris Capistrano
and Gerald G. Marten

Southeast Asia has a land area of 4.5 million square kilometers (km^2). About half the area is continental (Burma, Thailand, Vietnam, Laos, Cambodia, Singapore, and peninsular Malaysia), and the other half is archipelagic or insular (Indonesia, the Philippines, Brunei, Sabah, and Sarawak). Most of Southeast Asia has a tropical humid climate with tropical rainforest as the major form of natural vegetation. The Malay Peninsula, Borneo, Sumatra, West Java, the Moluccas, and the eastern Philippines receive abundant rainfall throughout the year. Areas such as peninsular Thailand, coastal Burma, Kampuchea, Sulawesi, and Mindanao have a definite dry season but sufficient rainfall (greater than 1,900 millimeters [mm] per year) that the soil seldom dries out. Most of the rest of Southeast Asia (e.g., central Burma, the central plain of Kampuchea, Northeast Thailand, and the leeside of mountains in Java, Sulawesi, and the Philippines) has a tropical subhumid climate with typically less than 1,500 mm of rainfall per year. Annual evapotranspiration in subhumid areas exceeds annual precipitation, so the soil dries out for part of the year.

Southeast Asia has a variety of soils that vary in both origin and fertility (Dudal et al., 1974). While there are highly weathered soils that are inherently low in fertility and highly vulnerable to leaching (e.g., in Malaysia and Sumatra), there are also soils (e.g., in Java) where the periodic fallout of volcanic ash has maintained high soil fertility. Heavy rainfall combined with agriculture on hillsides often has led to erosion that has reduced the agricultural productivity of upland areas, but deposition of sediment in the lowlands has created large areas of fertile alluvial soils that form the basis for wet-rice culture in the region. It also has been possible to extend rice production beyond the lowland areas through human efforts such as irrigating, terracing, and impounding waters on fields by bunds or retaining walls.

Agriculture is the principal economic activity in Southeast Asia (Duckham and Masefield 1969). Approximately 16 percent of the land area is under agriculture (Table 1.1). Agriculture provides employment to 61 percent of the total economically active population, and about 35 percent of the gross domestic product comes from agriculture (Table 1.2). Most of the agriculture

in the region is the kind of small-scale, traditional, subsistence agriculture that has prevailed in the region for millenia, though much of that agriculture has been changing under the impact of the Green Revolution. Commercial cash crops have become progressively more important as increasing numbers of farmers have been incorporated into national and international market economies directed toward the needs of expanding urban populations. As a consequence of the rapid population growth in the region in recent decades, intensive agriculture, which previously occurred primarily in lowland areas, has expanded significantly into upland areas that had been forested. Plantation agriculture based on crops such as coconut, sugar, rubber, tea, bananas, and abaca has been a prominent feature of much of the region for the past several centuries, with many Southeast Asian countries depending heavily upon the export of plantation crops to maintain their balance of trade.

Although traditional practices are still a major part of small-scale agriculture in Southeast Asia, these farmers are far from inertial. The agriculture they employ is a mix of practices passed to them by their parents and grandparents, crops they have observed to be successful on neighbors' fields, new varieties and technologies that have come to them through government development programs, and new inputs (e.g., mineral fertilizers and pesticides) that have come to them through the private sector. Most farmers experiment on a small scale with new crops and new cultivation techniques; they are willing to change their practices if convinced the changes will improve their incomes, and they often change their crops in response to changing market opportunities. Nonetheless, beneath these changes lie a sense of caution and prudence built upon a traditional technology that has endured and served them well over the centuries.

Table 1.3 presents a broad classification of contemporary agriculture in Southeast Asia. The significance of traditional technology in each kind of agriculture listed in Table 1.3 varies considerably. Subsistence oriented agricultural systems (e.g., swidden agriculture, rainfed rice, and other forms of upland agriculture) are still based primarily on traditional technologies. Commercially oriented agricultural systems (e.g., commercial vegetable gardens and annual and perennial plantation crops), as well as agricultural systems that have been the object of intensive development programs (e.g., wet rice), tend to have a much greater measure of modern Western technology.

One of the most conspicuous characteristics of traditional agricultural technology is the diversity of crops it employs. It is typical for a subsistence household to employ a number of cropping systems and a variety of crops within each of those cropping systems, including the interplanting of different crops in the same field. This crop diversity strongly shapes the ways traditional farmers perceive the natural resources available to them for agricultural production, structures their agriculture decisions, and influences how they organize their labor. Crop diversity also has a profound impact on how their agriculture performs, including not only the quantities of agricultural products they obtain but also the extent to which they can

Table 1.1. Land Use in Southeast Asia as of 1980
(in 000 hectares)

	Total Land Area	Annual Cropland[a]	Perennial Cropland[b]	Permanent Pasture[c]	Forest and Woodland[d]
Burma	65,774	9,573	450	361	32,167
Indonesia	181,135	14,200	5,300	12,000	121,800
Kampuchea	17,652	2,900	146	580	13,372
Laos	23,080	860	20	800	13,000
Malaysia	32,855	1,000	3,310	27	22,390
Philippines	29,817	7,050	2,870	1,000	12,300
Thailand	51,177	16,250	1,720	308	15,790
Vietnam	32,536	5,595	460	4,870	10,330
Southeast Asia Total	434,026	57,428	14,276	19,946	241,149

Source: FAO (1982).

[a] Land under temporary crops (double cropped areas are counted only once), temporary meadows for mowing pasture, land under market and kitchen gardens, and land temporarily fallow or lying idle.

[b] Land cultivated with crops that occupy the land for long periods and need not be replanted after each harvest, such as cocoa, coffee, and rubber; it includes land under shrubs, fruit trees, nut trees, and vine but excludes land under trees grown for wood or timber.

[c] Land in permanent (five years or more) herbaceous forage crops, either cultivated or growing wild (wild prairie or grazing land).

[d] Land under natural or planted stands of trees, productive or not, includes land from which forests have been cleared but will be reforested in the foreseeable future.

Table 1.2. The Significance of Agriculture in Southeast Asia

	Percent of Economically Active Population in Agriculture	Percentage of Gross Domestic Product Due to Agriculture
Burma	51	44
Indonesia	58	34
Kampuchea	74	–
Laos	73	–
Malaysia	47	30
Philippines	45	29
Thailand	75	30
Vietnam	70	–

Sources: IBRD-World Bank (1980), FAO (1982).

Table 1.3. A Broad Classification of Southeast Asian Agroecosystems

Type	Land Tenure	Unit of Production	Disposal of Output	Water Source
Rice-based agriculture				
Irrigated	Owned/leasehold/rental[b]	Household/corporation	Subsistence/market	Irrigated
Rainfed	Owned/communal (particularly uplands)	Household	Subsistence/market	Rainfed (sometimes tidal water)
Shifting cultivation	Communal/government land	Household/clan/settlement	Subsistence/market	Rainfed
Permanent field crops[a]	Owned/leasehold/rental[b]	Household	Subsistence/market	Rainfed/irrigated
Homegardens	Owned	Household	Subsistence/market	Rainfed (sometimes supplementary irrigation)
Commercial field crops				
Vegetables	Owned/government land	Household	Market	Mostly irrigated
Annual plantation or cash crops	Owned/leasehold/rental[b]	Household/corporation	Market/processing	Mostly irrigated
Perennial plantations	Owned/leasehold	Household/corporation	Market/processing	Irrigated/rainfed

Table 1.3. (*continued*)

Type	Crop Species Diversity	Degree of Modern Technology	Scale of Operation	Topography
Rice-based agriculture				
Irrigated	Rice monoculture, often rotated with other crops	Medium to high	Small to large	Lowlands (alluvial plains), terraced uplands
Rainfed	Rice monoculture, sometimes rotated with other crops	Low to medium	Small to medium	Upland rice in hilly areas, rainfed wet rice in lowlands including swamplands
Shifting cultivation	High	Low	Small	Wet tropical slope lands and upland areas
Permanent field crops[a]	Low to high	Low to medium	Small to medium	Flat to hilly, any cultivable lands not used for rice
Homegardens	High	Low to medium	Small	Slope lands or flat lands
Commercial field crops				
Vegetables	Low	High	Small to medium	Cool uplands
Annual plantation or cash crops	Monoculture	High	Medium to large	Flat lands
Perennial plantations	Low	Low to high	Small to large	Flat to hilly

[a] Other than rice.

[b] "Owned" signifies owner operated; leasehold and rental payments may be in cash or through a sharecropping arrangement.

Table 1.4. Area under Rice Cultivation in Southeast Asia

	Total Rice Area 000 ha	Percent		
		Rainfed Upland Rice[a]	Rainfed Wet Rice[b]	Irrigated Rice
Burma	5,500	0	84	16
Indonesia	9,005	16	45	39
Laos	739	29	57	14
Philippines	3,500	14	46	40
Thailand	9,140	10	66	24
Vietnam	5,615	8	79	13

Source: Modified from IRRI (1978).

[a] Unbounded rice.

[b] Including deep water rice.

count on it to meet their needs from year to year and to endure in the longer term.

RICE-BASED AGRICULTURE

Rice is generally the preferred food of Southeast Asia wherever conditions are suitable for its cultivation (Table 1.4). Wet rice is a form of permanent agriculture that dominates lowland areas, but it also occurs in some of the more densely populated upland areas, where terraces have been constructed to facilitate permanent agriculture. Wet rice can be grown throughout the year if there is year-round irrigation; where irrigation is not possible, and the rice is therefore "rainfed," it can be grown only during the rainy season because wet rice requires flood conditions. Not all rice requires flooding, however. "Upland rice" (also called "dry rice"), which is grown without flooding the field, is the most common form of rice in upland areas where flat land is scarce.

Private ownership of land is common among small-scale, irrigated rice cultivators in the lowlands, but many rice farmers also operate under leasehold or rental arrangements, particularly in areas where productivity is high. In some wetland rice areas, particularly in the uplands, the ownership of land is not legally recognized by the government, as, for example, among the Bontok, who cultivate wet rice on terraces in mountainous areas of northern Luzon in the Philippines (described in Chapter 3). Most wet-rice cultivation is small scale, with most of the produce being used for household subsistence and most of the remainder being used to pay rent to the landlord and cover expenses for agricultural inputs. The household is normally the unit of production, though exchange labor and wage labor may be employed during critical labor periods such as planting and harvesting, particularly if production is primarily for market rather than subsistence. Plowing is

traditionally by means of draft animals such as the buffalo. Small-scale tractor equipment also has been used in recent years; usually it is owned by one person in the village and rented to others. There is some large-scale commercial production, such as corporate rice farms in the Philippines and state farms in Burma, where tractors, rotary tillers, weeders, slashers, and other farm machinery are used along with hand labor.

Land preparation for wet rice consists of puddling, plowing, and harrowing. The rice may be seeded or transplanted into the fields. The use of commercial inorganic fertilizers and chemical pesticides is now widespread, particularly in the larger-scale operations. As a consequence of international programs to intensify rice production through nonphotoperiod-sensitive varieties and the expansion of year-round irrigation, continuous cropping of rice has become increasingly common in Southeast Asia. Although rice yields have increased significantly, these improvements have been accompanied by increasing hazards such as salinization and the onset of serious pest and disease problems that were not present under less intensive cultivation.

In areas where improved varieties are planted, it is common for all the paddies in a large area to be of the same variety. In more traditional areas there may be dozens of different rice varieties in a landscape mosaic of fields. Even a single household may employ several local varieties. In some areas it is customary to grow traditional varieties during the wet season for home consumption and improved varieties at other times of the year as a cash crop. Although rice is the overwhelmingly dominant crop in areas where it can be cultivated, it is also common to plant other crops such as bananas, cassava, yams, beans, or scattered fuelwood trees around the borders of rice fields. (Chapter 13 describes the roles of trees in rice fields). Rice is grown all year in many areas where conditions permit, but other crops such as vegetables may be grown in the paddies instead of rice during part of the year. This is common when irrigation is not available and vegetables can use residual soil moisture following rice cultivation. A mixture of vegetables may be planted in traditional cultivation, but more commercially oriented agriculture usually employs a monoculture of a single vegetable crop, as described in the section on commercial field crops later in this chapter. Animal products, particularly fish, can also be important in rice fields (Huat and Tan 1980). (Chapter 4 describes irrigated rice in Northern Thailand that is followed by a field-crop monoculture; Chapter 5 describes rainfed wet rice in Northeast Thailand that is followed by a field crop.)

Rainfed rice in upland areas tends to be different from the wet rice in lowland areas in a number of ways. Chemical inputs are used less and farm machinery is not so common in rainfed rice cultivation. Land tenancy is not so common with upland rice agriculture, because there usually is not sufficient surplus production from the upland rice to support anyone beyond those actually working the fields. Upland rice farmers usually work their own land, hire farm labor to a lesser extent, and consume their produce themselves. The presence of crops other than rice is much more conspicuous with upland rice than wet rice. (The highland section of Chapter 4 provides an example of upland rice cultivation.)

SHIFTING CULTIVATION

Shifting cultivation, also called swidden agriculture, is perhaps the most common form of agriculture for crops other than rice in Southeast Asia, particularly in upland areas but also in lowland areas where there is a low human population density and tropical forest still remains (Olofson 1981). Shifting cultivation is characterized by an alternation between crops and fallow. In a typical sequence, a forest is cut and then burned to clear the land and provide ash as "fertilizer" or "lime" for the soil. Crop yields are typically high for the first several years but subsequently decline due to problems such as excessive soil acidity, a decline in soil fertility, or invasion by weeds, insects, or other pests. As soon as yields decline, the field is abandoned as the farmer cuts another patch of forest to establish another swidden field nearby. Once abandoned, a field passes into fallow that may last for several years or several decades before the farmer returns to the same place, cuts the forest, and once again plants his crops. (This process is described in detail in Chapter 11).

Among the Lua tribe of northern Thailand (Kunstadter et al. 1978), dry rice is the primary crop, but there may be dozens of other crops mixed together in the same swidden field. (Shifting cultivation in this area is described in Chapter 4.) Several different strains of rice may be used to stagger the harvest in time to distribute labor requirements for harvesting over a longer period and to distribute the risk of crop losses in heavy storms that occur when the crop is particularly vulnerable immediately before harvest. Sorghum is planted along the borders of the field to mark the boundaries between fields, and a variety of crops such as chili peppers, root and tuber crops (taro, sweet potatoes, yams, cassava), creeping plants (squash, melons, cucumbers, gourds), and herbs and seasonings (garlic, onions, coriander, lemongrass, mint) are grown in patches or interspersed through the field. Maize and beans also may be planted in scattered patches. The crops are not planted neatly in rows and may appear to be planted at random, but their placement may in fact be attuned to micro-environmental variations within the field.

Shifting cultivation is often characterized by a progression of different crops from year to year. For example, the Hanunoo in the Philippines plant rice and maize during the first year after clearing, then root crops, such as sweet potatoes, yams, and cassava, and finally bananas, fruits, bamboo, and abaca after several years (Conklin 1957). The field eventually reverts to forest as the fruit trees age and forest trees invade the field. Trees may be planted in the field deliberately to provide fuelwood or building materials. (Shifting cultivation in mountainous terrain of the Philippines is described in Chapter 3.)

The tools for shifting cultivation are simple. Knives and axes of various sizes and shapes are used for clearing, cutting, and harvesting. Land preparation may simply be a matter of punching holes in the soil with a dibble stick before dropping in seeds.

Land ownership is usually communal or tribal among aboriginal shifting cultivators, but some form of private property usually exists among peasants. The household is the usual unit of production, but the clan or settlement can also provide a basis for sharing labor between households. The land tenure of shifting cultivators can be ambiguous when they operate on forested lands that are considered to fall within government domain.

A variety of changes have occurred in shifting cultivation over the past several centuries. For example, cash crops such as pepper and rubber have been added by the colonial economy. In some cases this happened on a small scale, but in other cases large areas of land were cleared by commercial enterprises or government agencies to establish plantations of rubber, pepper, cassava, coconuts, coffee, abaca, teak, or tea with shifting cultivators doing the planting. There has also been considerable migration of lowland (non-swidden) farmers to upland areas where they have had to practice shifting cultivation (Chapman 1974). As population densities have increased, the amount of land per household has decreased correspondingly, and there has been a trend toward shorter fallow periods with a corresponding reduction in the ecological benefits provided by the forest fallow.

The end product of this trend is a permanent field with an interplanted mixture of field crops and scattered perennials such as bananas, citrus trees, or other fruit trees. Production is primarily for subsistence, but a substantial quantity of the produce may also be sold. This form of agriculture is usually rainfed but may extend through most of the year in the more humid parts of Southeast Asia. (A typical example is the Javanese *kebun* described in Chapters 6 and 14.) It is typical for a field to contain a grain crop such as maize, sorghum or millet, and a variety of legumes such as peanuts, cowpeas, soybeans, or mung beans. Legumes have the advantage that they tend to be drought tolerant, reach maturity rapidly, fix atmospheric nitrogen, have a high protein and calorie content, are easily stored, and have a ready market.

HOMEGARDENS

Homegardens in Southeast Asia typically occupy an area of about 1,000 square meters or less immediately surrounding the house. They vary greatly with climate and local custom. Some, particularly in more urban areas where there is little land around the house, may consist entirely of a kitchen garden similar to the American concept of a vegetable garden, or they may be dominated by ornamental plants. Perennials such as fruit trees are a conspicuous feature of most homegardens in Southeast Asia, however, and in some of the more humid areas (e.g., Malaysia, Java, and Bali), there are so many different kinds of trees and field crops in the homegarden, and they are growing in such abundance, that it looks more like a tropical forest than a garden. (See Chapters 6 and 14.) Even the wildest looking homegarden is usually in reality a collection of domesticated and semidomesticated plants with a variety of uses including food, fuel, construction

Table 1.5. Common Commercial Vegetable Crops

Botanical Name	English Name
Allium ascalonicum	Shallot
Allium cepa	Onion
Allium sativum	Garlic
Apium graveolens	Chinese celery
Brassica albograba	Chinese kale
Brassica chinensis	Chinese cabbage
Brassica oleracea	Cabbage, cauliflower, brussels sprout
Capsicum annuum	Chili
C. frutescens	Bird pepper
Coriandrum sativum	Coriander
Cucumis sativus	Cucumber
Ipomea batatas	Sweet potato
Lactuca sativa	Lettuce
Momordica charantia	Bitter gourd
Nasturtium aquaticum	Chinese cress
Solanum melongena	Eggplant
Vigna sinensis	Bean sprout

Source: Hill (1982).

materials, herbal medicine, ornamentation, and shade (Terra 1958). Homegardens are often in continuous production throughout the year and lend themselves to intensive care because they are so conveniently close to the house. They can be fertilized with kitchen wastes, receive supplementary irrigation with well water, and be attended by women and children in their spare time.

COMMERCIAL FIELD CROPS

Vegetables

Commercial vegetable gardening has expanded rapidly in recent decades in response to increasing urban markets for potatoes and vegetables like those listed in Table 1.5. Vegetable gardening in Southeast Asia often has been associated with Chinese immigrants, who traditionally maintained vegetable gardens in association with raising pigs (Hill 1982). Other ethnic groups have taken up this kind of agriculture after acquiring the technology from the Chinese, but the Chinese have continued to play a major role as middlemen in the marketing process. Commercial vegetable gardening has become particularly prominent in cool upland areas that are suitable for temperate vegetables, such as the Cameron highlands in peninsular

Malaysia, Mountain Province of northern Luzon in the Philippines (see Chapter 3), and the hills of Chiangmai region in Thailand (see Chapter 4). These areas generally are restricted to locations where there are roads to carry the products out to market.

The basic unit of production is usually the household, but part-time or full-time laborers may be employed. Some lands are privately owned, others are under communal or tribal ownership, and others are located illegally on government forest lands. Commercial vegetable gardening is intensive, typically involving five to ten crops per year. The time interval between the harvest of one crop and formation of a new bed for the next crop is rarely more than a week or two, seedlings are transplanted from nursery beds, and weeding is continuous. Garden plots are usually monoculture and large quantities of chemical inputs are employed, a distinct contrast to the mixed cropping, self-sufficient style of traditional subsistence gardening in Southeast Asia. A continuous water supply is essential for commercial vegetable gardening to assure year-round cultivation.

Annual Plantation or Cash Crops

Vegetable gardens are not the only form of commercial field crops. Annual plantation crops such as tobacco, cotton, sugarcane, indigo, pineapple, and bananas are often grown in monocultures for export or local processing. (Sugarcane production of this sort is described in Chapter 3.) Care is relatively intensive and chemical inputs are high. Whereas plantations of perennial export crops are usually managed on a large scale by government corporations or multinational firms, the plantation production of annual crops is more commonly on small landholdings through contract agreements with local processing plants or marketing agencies. Small holders provide the land and labor while the company furnishes seeds, technical expertise, an assured market (provided the product meets company standards), and often credit.

PERENNIAL AGROECOSYSTEMS

Small Holders

High-value crops such as coconuts, rubber, citrus, and other fruit trees are grown primarily for sale. Small-scale farmers may have some fruit trees near their houses where they can keep an eye on them, or they may have a larger patch of trees (e.g., rubber) in a more upland area, sometimes a considerable distance from the house. The level of chemical inputs and degree of modern agricultural technology used is greater than in subsistence agriculture but usually less than in large-scale perennial crop plantations.

While some perennial crops (e.g., rubber and tea) do not lend themselves to intercropping, it is common to have additional crops planted between coconut or fruit trees, particularly when land is scarce and plot sizes are small and the land must be used intensively. During the first few years after

Table 1.6. Areas Planted to Rubber and Coconut Plantations in 1971 (000 hectares)

	Rubber	Coconuts
Burma	62	–
Cambodia	48	–
Indonesia	1,987	1,200
Malaysia	2,006	532
Philippines	22	2,125
South Vietnam	100	–
Thailand	735	165

Sources: Fryer (1979)

a perennial crop has been planted, there is considerable space between the seedlings, making it possible to interplant annual crops that require direct sunlight. Annual crops can provide an income for the household at a time when the perennials are not yet producing, and the fertilization and other care provided to annual crops can be beneficial to the perennial crops as well, increasing coconut yields by as much as 50 percent (Gomez and Gomez 1983), for example. As a perennial crop matures and its canopy closes over, it is no longer feasible to interplant most annual crops, and more shade-tolerant crops (e.g., bananas, pineapple, coffee, or cacao) are interplanted. The tenancy of small-scale perennial crop production varies considerably, but there is a particular incentive for interplanting where a share of the major crop (e.g., coconut) must be given to the landlord while the cultivator can keep the produce from interplanted crops (Poffenberger 1983). Interplantings, particularly in coconut plantations, may also include leguminous grasses for grazing cattle.

Large-Scale Plantations

Large-scale agricultural plantations were established in Southeast Asia by Europeans during the colonial period. Rubber has been the most important plantation crop (Table 1.6) and still occupies half the large-scale plantation land in Southeast Asia, though most of it (73 percent of the area planted to rubber) is cultivated by small holders (Fryer 1979). Coconut, cacao, oil palm, and tea also are significant plantation crops. Perennial crop plantations are prominent in the humid part of Southeast Asia where the climate permits continuous, year-round harvesting, thereby facilitating an even deployment of labor and equipment. Crops with distinctly seasonal yields tend to fall within the domain of small-scale producers. Since processing the product may be rather complex (Grigg 1974), plantations often are located along navigable rivers and coasts to facilitate the transport of inputs and outputs. Plantations tend to be particularly significant in countries with a colonial past (e.g., Malaysia, Indonesia, Kampuchea, and Vietnam).

The primary unit of production in plantation agriculture is the company or corporation, and management decisions rest with professionals who do not themselves work in the plantations (Blaut 1961). Plantation lands generally are owned by the company or leased from the government. There is extensive use of modern agricultural technology, mechanization, and chemical inputs, and hired laborers are employed for tasks that cannot be handled by machines. There has been considerable plantation crop research—including the development of erosion-control measures such as terracing, ditching, contour planting, surface-cover vegetation, shade trees, and hedges—because of the economic interests that colonial governments had in this kind of agriculture and the long-term investments of corporations in their plantations. Except for crops such as coffee and cacao, which may require shade trees, large-scale perennial plantations are usually monocultures.

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REFERENCES

Blaut, J. 1961. The ecology of tropical farming systems. *Revista Geografica* 54:47–67.

Chapman, E.C. 1974. Shifting cultivation in tropical forest areas of Southeast Asia. Paper presented at the Regional Meeting of International Union for the Conservation of Nature, May–June 1974, Bandung, Indonesia.

Conklin, H.C. 1957. Hanunoo agriculture: A report on an integral system of shifting cultivation in the Philippines. Forestry Development Papers 12, FAO, Rome.

Duckham, A.N., and G.B. Masefield. 1969. *Farming Systems of the World.* New York, N.Y.: Praeger.

Dudal, R., F. Moorman, and J. Riquier. 1974. Soils of humid tropical Asia. In *Natural Resources of Humid Tropical Asia*, pp. 159–78. Natural Resources Research, XII, UNESCO, Paris.

FAO. 1982. *Production Yearbook* 1981, vol. 35. Rome.

Fryer, D. 1979. *Emerging Southeast Asia: Study on Growth and Stagnation.* New York, N.Y.: John Wiley and Sons.

Gomez, A.A., and K.A. Gomez. 1983. *Multiple Cropping in the Humid Tropics.* Ottawa, Canada: International Development Research Center.

Grigg, D.B. 1974. *The Agricultural Systems of the World: An Evolutionary Approach.* London: Cambridge University Press.

Hill, R.D. 1982. *Agriculture in the Malaysian Region.* Budapest, Hungary: Akademiai Kiado.

Huat, K.K., and E.S.P. Tan. 1980. Review of rice-fish culture in Southeast Asia. In *Integrated Agriculture—Aquaculture Farming Systems*, eds. R.S.V. Pullin and Z.H. Shehadeh, pp. 1–14. International Center for Living Aquatic Resources Management, Manila, and Southeast Asian Regional Center for Study and Research in Agriculture, Los Banos, Philippines.

IBRD-World Bank. 1980. *World Tables.* 2nd ed. Baltimore: The Johns Hopkins University Press.

IRRI. 1978. World rice statistics. International Rice Research Institute, Los Banos, Philippines.

Kunstadter, P., E.C. Chapman, and S. Sabhasri, eds. 1978. *Farmers in the Forest: Economic Development and Marginal Agriculture in Northern Thailand.* Honolulu: University Press of Hawaii.

Olofson, H., ed. 1981. Adaptive strategies and change in Philippine swidden-based societies. Forest Research Institute, University of the Philippines, Los Banos.

Poffenberger, M. 1983. Changing dryland agriculture in eastern Bali. *Human Ecology* 11:123–44.

Terra, G.J.A. 1958. Farm systems in Southeast Asia. *Netherlands Journal of Agricultural Science* 6:157–82.